

FRP precast grid panel with precast square gratings measuring 1½ "×1½" in size, height 1", 70% elasticity, and safety factor 5:1

Values = deflection (in) at applied load Safe Load = max safe load (lbs/ft² or lbs/ft) — = beyond usable range

Span (in)	Type	50 psf	100 psf	150 psf	200 psf	250 psf	300 psf	400 psf
12	U (Uniform)	—	—	0.013	0.017	0.021	0.025	0.034
	C (Conc.)	—	0.014	0.020	0.027	0.034	0.041	0.054
18	U (Uniform)	0.021	0.041	0.062	0.082	0.103	0.123	0.164
	C (Conc.)	0.022	0.044	0.066	0.088	0.110	0.131	0.175
24	U (Uniform)	0.064	0.128	0.192	0.256	0.320	0.384	0.512
	C (Conc.)	0.051	0.102	0.154	0.205	0.256	0.307	0.409
30	U (Uniform)	0.155	0.309	0.464	0.619	—	—	—
	C (Conc.)	0.099	0.198	0.297	0.396	0.495	0.594	—
36	U (Uniform)	0.318	0.635	—	—	—	—	—
	C (Conc.)	0.169	0.339	0.508	0.677	—	—	—

FRP precast grid panel, precast square grating 1½ "×1½" with height H=1½", 70% perforati

Values = deflection (in) at applied load Safe Load = max safe load (lbs/ft² or lbs/ft) — = beyond usable range

Span (in)	Type	50 psf	100 psf	150 psf	200 psf	250 psf	300 psf	400 psf
12	U (Uniform)	—	—	—	—	—	0.011	0.014
	C (Conc.)	—	—	—	0.011	0.014	0.017	0.023
18	U (Uniform)	—	—	0.021	0.028	0.036	0.043	0.057
	C (Conc.)	—	0.015	0.023	0.030	0.038	0.046	0.061
24	U (Uniform)	0.021	0.042	0.063	0.084	0.104	0.125	0.167
	C (Conc.)	0.017	0.033	0.050	0.067	0.084	0.100	0.134
30	U (Uniform)	0.047	0.094	0.141	0.188	0.235	0.283	0.377
	C (Conc.)	0.030	0.060	0.090	0.121	0.151	0.181	0.241
36	U (Uniform)	0.096	0.192	0.288	0.384	0.480	0.576	—
	C (Conc.)	0.051	0.102	0.154	0.205	0.256	0.307	0.410
42	U (Uniform)	0.175	0.350	0.525	—	—	—	—
	C (Conc.)	0.080	0.160	0.240	0.320	0.400	0.480	0.641
48	U (Uniform)	0.287	0.573	—	—	—	—	—
	C (Conc.)	0.115	0.229	0.344	0.459	0.573	0.688	—

FRP precast grating panel, precast grating 2 "×2", H=2", 72% opening rate, 4.0

Values = deflection (in) at applied load Safe Load = max safe load (lbs/ft² or lbs/ft) — = beyond usable range

Span (in)	Type	50 psf	100 psf	150 psf	200 psf	250 psf	300 psf	400 psf
12	U (Uniform)	—	—	—	—	—	—	—
	C (Conc.)	—	—	—	—	—	0.010	0.013
18	U (Uniform)	—	—	0.012	0.016	0.020	0.024	0.032
	C (Conc.)	—	—	0.013	0.017	0.021	0.026	0.034
24	U (Uniform)	0.010	0.021	0.031	0.042	0.052	0.063	0.083
	C (Conc.)	—	0.017	0.025	0.033	0.042	0.050	0.067
30	U (Uniform)	0.023	0.046	0.069	0.092	0.114	0.137	0.183
	C (Conc.)	0.015	0.029	0.044	0.059	0.073	0.088	0.117
36	U (Uniform)	0.044	0.089	0.133	0.177	0.222	0.266	0.355
	C (Conc.)	0.024	0.047	0.071	0.095	0.118	0.142	0.189
42	U (Uniform)	0.082	0.164	0.245	0.327	0.409	0.491	0.654
	C (Conc.)	0.037	0.075	0.112	0.150	0.187	0.224	0.299
48	U (Uniform)	0.135	0.270	0.405	0.541	—	—	—
	C (Conc.)	0.054	0.108	0.162	0.216	0.270	0.324	0.432
54	U (Uniform)	0.210	0.420	0.630	—	—	—	—
	C (Conc.)	0.075	0.149	0.224	0.298	0.373	0.448	0.597

FRP precast grating, precast rectangular 1 "x4", H=1", 69% perforated, 2.80 lb

Values = deflection (in) at applied load Safe Load = max safe load (lbs/ft² or lbs/ft) — = beyond usable range

Span (in)	Type	50 psf	100 psf	150 psf	200 psf	250 psf	300 psf	400 psf
12	U (Uniform)	—	—	—	0.011	0.014	0.017	0.022
	C (Conc.)	—	—	0.013	0.018	0.022	0.027	0.035
18	U (Uniform)	0.012	0.025	0.037	0.049	0.062	0.074	0.099
	C (Conc.)	0.013	0.026	0.039	0.053	0.066	0.079	0.105
24	U (Uniform)	0.037	0.074	0.112	0.149	0.186	0.223	0.298
	C (Conc.)	0.030	0.060	0.089	0.119	0.149	0.179	0.238
30	U (Uniform)	0.088	0.176	0.264	0.352	0.440	0.528	—
	C (Conc.)	0.056	0.113	0.169	0.225	0.282	0.338	0.451
36	U (Uniform)	0.176	0.353	0.529	—	—	—	—
	C (Conc.)	0.094	0.188	0.282	0.376	0.470	0.564	—
42	U (Uniform)	0.316	0.632	—	—	—	—	—
	C (Conc.)	0.144	0.289	0.433	0.577	—	—	—

FRP precast grating, precast rectangular with dimensions of 1½"×6", height H=1½", 67% perfor

Values = deflection (in) at applied load Safe Load = max safe load (lbs/ft² or lbs/ft) — = beyond usable range

Span (in)	Type	50 psf	100 psf	150 psf	200 psf	250 psf	300 psf	400 psf
12	U (Uniform)	—	—	—	—	0.011	0.014	0.018
	C (Conc.)	—	—	0.011	0.015	0.018	0.022	0.029
18	U (Uniform)	—	0.018	0.027	0.035	0.044	0.053	0.071
	C (Conc.)	0.010	0.019	0.028	0.038	0.047	0.057	0.076
24	U (Uniform)	0.019	0.038	0.056	0.075	0.094	0.112	0.150
	C (Conc.)	0.015	0.030	0.045	0.060	0.075	0.090	0.120
30	U (Uniform)	0.039	0.078	0.117	0.156	0.195	0.233	0.311
	C (Conc.)	0.025	0.050	0.075	0.100	0.125	0.150	0.200
36	U (Uniform)	0.071	0.143	0.214	0.285	0.357	0.428	—
	C (Conc.)	0.038	0.076	0.114	0.152	0.190	0.228	0.304
42	U (Uniform)	0.126	0.252	0.378	0.504	0.630	—	—
	C (Conc.)	0.058	0.115	0.173	0.230	0.288	0.346	0.461
48	U (Uniform)	0.207	0.414	0.621	—	—	—	—
	C (Conc.)	0.083	0.160	0.248	0.331	0.414	0.497	—

FRP pultruded Type I grid panels / pultruded Type I flat steel bars (1"×0.600", 6" ba

Values = deflection (in) at applied load Safe Load = max safe load (lbs/ft² or lbs/ft) — = beyond usable range

Span (in)	Type	100	200	300	400	500	750	1000
12	U (Uniform)	0.002	0.004	0.005	0.007	0.009	0.014	0.018
	C (Conc.)	0.003	0.006	0.009	0.012	0.015	0.022	0.029
18	U (Uniform)	0.008	0.017	0.025	0.033	0.042	0.063	0.084
	C (Conc.)	0.009	0.018	0.027	0.036	0.045	0.067	0.089
24	U (Uniform)	0.025	0.050	0.075	0.100	0.124	0.187	0.249
	C (Conc.)	0.020	0.040	0.060	0.080	0.100	0.149	0.199
30	U (Uniform)	0.058	0.116	0.174	0.231	0.289	0.434	0.579
	C (Conc.)	0.037	0.074	0.111	0.148	0.185	0.278	0.370
36	U (Uniform)	0.115	0.230	0.345	0.460	0.575	—	—
	C (Conc.)	0.061	0.123	0.184	0.245	0.307	0.460	0.614
42	U (Uniform)	0.211	0.422	0.633	—	—	—	—
	C (Conc.)	0.096	0.193	0.289	0.386	0.482	—	—
48	U (Uniform)	0.353	0.705	—	—	—	—	—

	C (Conc.)	0.141	0.282	0.423	0.564	—	—	—
54	U (Uniform)	0.563	—	—	—	—	—	—
	C (Conc.)	0.200	0.400	0.600	—	—	—	—

FRP pultruded I-beam – MS-I-6015 pultruded I-shaped flat steel, dimensions 1½ "×0.600

Values = deflection (in) at applied load Safe Load = max safe load (lbs/ft² or lbs/ft) — = beyond usable range

Span (in)	Type	100	200	300	400	500	750	1000
12	U (Uniform)	0.001	0.001	0.002	0.003	0.003	0.005	0.006
	C (Conc.)	0.001	0.002	0.003	0.004	0.005	0.008	0.010
18	U (Uniform)	0.003	0.006	0.009	0.012	0.015	0.023	0.030
	C (Conc.)	0.003	0.006	0.010	0.013	0.016	0.024	0.032
24	U (Uniform)	0.009	0.018	0.027	0.037	0.046	0.069	0.091
	C (Conc.)	0.007	0.015	0.022	0.029	0.037	0.055	0.073
30	U (Uniform)	0.022	0.043	0.065	0.086	0.108	0.161	0.215
	C (Conc.)	0.014	0.028	0.041	0.055	0.069	0.103	0.138
36	U (Uniform)	0.044	0.087	0.131	0.175	0.218	0.327	0.436
	C (Conc.)	0.023	0.047	0.070	0.093	0.116	0.175	0.233
42	U (Uniform)	0.079	0.159	0.238	0.317	0.396	0.595	—
	C (Conc.)	0.036	0.072	0.109	0.145	0.181	0.272	0.362
48	U (Uniform)	0.133	0.266	0.400	0.533	0.666	—	—
	C (Conc.)	0.053	0.107	0.160	0.213	0.266	0.400	0.533
54	U (Uniform)	0.211	0.422	0.633	—	—	—	—
	C (Conc.)	0.075	0.150	0.225	0.300	0.375	0.563	—

FRP pultruded Type I grid – MS-I-6515 pultruded Type I flat steel, dimensions 1½ "×0.600", 8"

Values = deflection (in) at applied load Safe Load = max safe load (lbs/ft² or lbs/ft) — = beyond usable range

Span (in)	Type	100	200	300	400	500	750	1000
12	U (Uniform)	0.001	0.001	0.002	0.003	0.004	0.005	0.007
	C (Conc.)	0.001	0.002	0.004	0.005	0.006	0.009	0.012
18	U (Uniform)	0.003	0.007	0.010	0.014	0.017	0.026	0.035
	C (Conc.)	0.004	0.007	0.011	0.015	0.018	0.028	0.037
24	U (Uniform)	0.010	0.021	0.031	0.042	0.052	0.078	0.104
	C (Conc.)	0.008	0.017	0.025	0.033	0.042	0.063	0.083
30	U (Uniform)	0.025	0.049	0.074	0.098	0.123	0.184	0.245

	C (Conc.)	0.016	0.031	0.047	0.063	0.079	0.118	0.157
36	U (Uniform)	0.050	0.100	0.149	0.199	0.249	0.373	0.498
	C (Conc.)	0.027	0.053	0.080	0.106	0.133	0.199	0.265
42	U (Uniform)	0.090	0.181	0.271	0.361	0.452	0.678	—
	C (Conc.)	0.041	0.083	0.124	0.165	0.207	0.310	0.413
48	U (Uniform)	0.152	0.304	0.456	0.607	—	—	—
	C (Conc.)	0.061	0.121	0.182	0.243	0.304	0.456	0.607
54	U (Uniform)	0.241	0.481	—	—	—	—	—
	C (Conc.)	0.086	0.171	0.257	0.342	0.428	0.642	—
60	U (Uniform)	0.364	—	—	—	—	—	—
	C (Conc.)	0.117	0.233	0.350	0.467	0.583	—	—
66	U (Uniform)	0.531	—	—	—	—	—	—
	C (Conc.)	0.155	0.309	0.464	0.618	—	—	—

% perforation rate, 2.6 lbs/ft² modulus of

500 psf	Safe Load lbs/ft ²	Safe Load lbs/ft
0.042	1,360	
0.068		680

0.205	666	
0.219		500

0.640	380	
0.512		380

—	240	
—		300

—	160	
—		240

on rate, 3.8 lbs/ft², safety factor 5:1

500 psf	Safe Load	Safe Load
0.018	3,120	
0.028		1,560

0.071	1,386	
0.076		1,040

0.209	780	
0.167		780

0.471	496	
0.301		620

—	347	
0.512		520

—	251	
0.801		440

—	170	
—		340

lbs/ft², safety factor 5:1

500 psf	Safe Load	Safe Load
0.010	4,000	
0.016		2,000

0.040	1,813	
0.043		1,360

0.104	960	
0.083		960

0.229	640	
0.146		800

0.444	453	
0.237		680

—	331	
0.374		580

—	260	
0.541		520

—	204	
—		460

s/ft², safety factor 5:1

500 psf	Safe Load	Safe Load
0.028	1,960	
0.044		980

0.123	960	
0.131		720

0.372	560	
0.298		560

—	336	
0.563		420

—	240	
—		360

—	183	
—		320

ation, weight 3.75 lbs/ft², safety factor 5:1

500 psf	Safe Load	Safe Load
0.023	4,272	
0.037		2,136

0.089	1,712	
0.095		1,284

0.188	956	
0.150		956

0.389	587	
0.250		734

—	385	
0.381		578

—	370	
—		472

—	184	
—		368

r spacing, 60% perforation rate, 2.4 lbs/ft², safety factor 2:1)

2000	3000	4000	5000	Safe Load	Safe Load
0.036	0.054	0.073	0.091	10,401	
0.058	0.087	0.116	0.145		5,200

0.167	0.251	0.335	0.418	4,954	
0.179	0.268	0.357	0.446		3,716

0.498	—	—	—	2,900	
0.398	0.597	—	—		2,900

—	—	—	—	1,856	
—	—	—	—		2,320

—	—	—	—	1,289	
—	—	—	—		1,933

—	—	—	—	943	
—	—	—	—		1,649

—	—	—	—	719	
---	---	---	---	-----	--

—	—	—	—	1,437
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—	—	—	—	566	
—	—	—	—		1,274

", 6" bar spacing, 60% opening ratio, 3.0 lbs/ft², safety factor 2:1

2000	3000	4000	5000	Safe Load	Safe Load
0.013	0.019	0.026	0.032	17,601	
0.020	0.031	0.041	0.051		8,800

0.061	0.091	0.121	0.152	7,823	
0.065	0.097	0.129	0.162		5,867

0.183	0.274	0.366	0.457	4,400	
0.146	0.220	0.293	0.366		4,400

0.430	0.646	—	—	2,773	
0.276	0.413	0.551	—		3,467

—	—	—	—	1,896	
0.466	—	—	—		2,845

—	—	—	—	1,361	
—	—	—	—		2,381

—	—	—	—	1,017	
—	—	—	—		2,033

—	—	—	—	777	
—	—	—	—		1,748

crossbar spacing, DURADEK material, 65% perforation rate, 2.7 lbs/ft²,

2000	3000	4000	5000	Safe Load	Safe Load
0.015	0.022	0.029	0.038	15,439	
0.023	0.035	0.047	0.058		7,719

0.069	0.091	0.104	0.138	6,862	
0.074	0.111	0.147	0.184		5,146

0.209	0.313	0.417	0.522	3,860	
0.167	0.250	0.334	0.417		3,860

0.491	—	—	—	2,433	
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0.314	0.471	0.628	—		3,041
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—	—	—	—	1,663	
—	—	—	—		2,495

—	—	—	—	1,194	
—	—	—	—		2,088

—	—	—	—	892	
—	—	—	—		1,784

—	—	—	—	681	
—	—	—	—		1,533

—	—	—	—	533	
—	—	—	—		1,333

—	—	—	—	425	
—	—	—	—		1,170

--	--	--	--	--	--